

2.4 Nitrogen & Sulfur

Question Paper

Course	CIEA Level Chemistry
Section	2. Inorganic Chemistry
Topic	2.4 Nitrogen & Sulfur
Difficulty	Easy

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

Extreme conditions are required to synthesise ammonia from nitrogen in the Haber process due to the lack of reactivity of the nitrogen molecule.

Give **two** reasons for the lack of reactivity of the nitrogen molecule, N_2 .

[2 marks]

Question 1b

Under conditions of high temperature, nitrogen and oxygen react together to give oxides of nitrogen.

i)

Write an equation for a possible reaction between nitrogen and oxygen.

[1]

ii)

State **two** situations, one natural and one as a result of human activities, in which nitrogen and oxygen react together.

[2]

iii)

State the **two** main environmental effects of the presence of nitrogen oxides in the atmosphere.

[2]

[5 marks]

Question 1c

i)

State how nitrogen monoxide is removed from the exhaust gases of motor vehicles.

[1]

ii)

Write an equation for this process.

[1]

[1 mark]**Question 2a**

Nitrogen oxides can react with substances in the air to make secondary pollutants.

One of these secondary pollutants is peroxyacetyl nitrate, commonly known as PAN.

i)

State the type of compounds that react with nitrogen oxides to form PAN.

[1]

ii)

PAN is one of the components found in photochemical smog. Explain why the term photochemical is used to describe smog.

[1]

[2 marks]**Question 2b**

One of the main reasons for reducing the amounts of oxides of nitrogen in the atmosphere is their contribution to the formation of acid rain.

i)

Write an equation for the formation of nitric acid from nitrogen dioxide, NO_2 , in the atmosphere.

[2]

ii)

State the role that nitrogen dioxide, NO_2 , plays in the oxidation of atmospheric sulfur dioxide, SO_2 , to form acid rain.

[1]

[3 marks]

Question 2c

State the oxidation number of nitrogen in nitrogen monoxide, NO, and nitrogen dioxide, NO₂.

oxidation number in NO

oxidation number in NO₂

[1 mark]

Question 3a

Ammonium sulfate is a fertiliser which is manufactured by the reaction between ammonia and sulfuric acid.

Ammonia is described as a weak base and sulfuric acid as a strong acid. By using an equation, explain clearly why ammonia can be described as a weak base.

[3 marks]

Question 3b

i)

Write the equation for the neutralisation of aqueous ammonia by dilute sulfuric acid.

[1]

ii)

State the bond angles between N-H bonds in ammonia and the ammonium ion. Explain any differences

[3]

[4 marks]**Question 3c**

Ammonia can be displaced from an ammonium salt by an acid–base reaction. One such reaction is between ammonium chloride and calcium hydroxide.

i)

Write the equation for the reaction between ammonium chloride and calcium hydroxide.

[2]

ii)

Explain how the presence of ammonia gas can be confirmed.

[1]**[3 marks]**

Model Answers: Easy

1a

a) The reason for the lack of reactivity of the nitrogen molecule is:

- (Due to the) strong $\text{N}\equiv\text{N}$ / triple bond; [1 mark]
- N_2 is a non-polar molecule; [1 mark]

[Total: 2 marks]

- The $\text{N}\equiv\text{N}$ triple bond has a high bond enthalpy meaning that a large amount of energy is required to break the bond
 - As it is so difficult to break, the N_2 molecule is very unreactive and requires extreme conditions to break the bond and allow a nitrogen atom to react
- Electrons are shared equally between the two nitrogen atoms so nitrogen molecules are non-polar
 - This lack of polarity means that nitrogen molecules are not likely to attract or react with other molecules

1b

b)

i) An equation for a possible reaction between nitrogen and oxygen is:



OR



ii) **Two** situations, one natural and one as a result of human activities, in which nitrogen and oxygen react together are:

- Natural = In lightning; [1 mark]
- Human activity = In an engine / combustion of fuels (or a specific example); [1 mark]

iii) **Two** main environmental effects of the presence of nitrogen oxides in the atmosphere are:

- Formation of photochemical smog; [1 mark]
- Production of acid rain; [1 mark]

[Total: 5 marks]

- Any balanced equation that forms a stable nitrogen oxide would be accepted for part (i)
- For example, the formation of N_2O , nitrous oxide:
 - $2\text{N}_2 + \text{O}_2 \rightarrow 2\text{N}_2\text{O}$
- You should know the equations for the formation of nitrogen monoxide, NO , and nitrogen dioxide, NO_2
- Due to the lack of reactivity of oxygen, nitrogen and oxygen present in the atmosphere will only react under extreme conditions
- Lightning is a natural situation that causes nitrogen and oxygen to react
- The high temperatures and pressures inside car engines (and aeroplane engines) are also extreme enough to cause nitrogen and oxygen to react
- The examples of how oxides of nitrogen are formed given in the mark scheme are the situations that you need to know, but there are other situations where oxides of nitrogen are formed that you would be given credit for, for example:
 - Volcanoes
 - Biological decay
 - Combustion of coal / oil at electric power plants
 - Manufacture of fertilisers

1c

c)

i) The NO_2 is removed from the exhaust gases of motor vehicles by:

- A catalytic converter

OR

Passing the exhaust gases over a (hot) catalyst / Pt / Rh; [1 mark]

ii) The equation for this process is:

- $2\text{NO} + 2\text{CO} \rightarrow \text{N}_2 + 2\text{CO}_2$; [1 mark]

[Total: 2 marks]

- Catalytic converters remove carbon monoxide, nitrogen oxide and nitrogen dioxide from exhaust gases by passing them over a hot catalyst which has a honeycombed structure to increase the surface area
- For nitrogen dioxide:
 - $\text{NO}_2 + 2\text{CO} \rightarrow \frac{1}{2}\text{N}_2 + 2\text{CO}_2$
- Both nitrogen monoxide and nitrogen dioxide are reduced

2a

a)

i) The type of compounds that react with nitrogen oxides to form PAN are:

- Unburnt hydrocarbons (from fuel)

OR

VOCs / volatile organic compounds; [1 mark]

ii) The term photochemical is used to describe smog because:

- Sunlight provides the energy to start the reaction (between nitrogen oxides and unburnt hydrocarbons / VOCs); [1 mark]

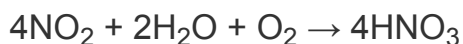
[Total: 2 marks]

- Nitrogen oxides and unburnt hydrocarbons are both primary pollutants as they are both given off directly into the air from the source of pollution
- Volatile organic compounds, known as VOCs, are mainly made up of unburnt hydrocarbons and the products of oxidation of these hydrocarbons
- VOCs will react with nitrogen oxides in the presence of sunlight to produce peroxyacetyl nitrate or PAN (formula $\text{CH}_3\text{CO}_3\text{NO}_2$) which is a component of photochemical smog
- PAN is a secondary pollutant as it is a pollutant that is not given off directly into the air from human activity
- Photochemical smog tends to be worst in cities in the afternoons of sunny days because:
 - There are a lot of cars emitting VOCs and nitrogen oxides
 - These build up during the day
 - The sunlight provides the necessary energy to start the reaction

2b

b)

i) An equation for the formation of nitric acid from nitrogen dioxide, NO_2 , in the atmosphere is:



- Correct formulae; [1 mark]
- Correct balancing; [1 mark]

ii) The role that nitrogen dioxide, NO_2 , plays in the oxidation of atmospheric sulfur dioxide, SO_2 , to form acid rain is:

- Catalyst; [1 mark]

[Total: 3 marks]

- When SO_2 , another atmospheric pollutant, is oxidised, it forms SO_3 which reacts with rainwater to form dilute sulfuric acid:
 - $\text{SO}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_4(\text{aq})$
- NO_2 catalyses the oxidation of SO_2 to SO_3 :
 - $\text{NO}_2(\text{g}) + \text{SO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$
- The formed NO gets oxidised to regenerate NO_2 :
 - $\text{NO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$

2c

c) The oxidation number of nitrogen in nitrogen monoxide, NO , and nitrogen dioxide, NO_2 are:

- +2 in NO ; [1 mark]
- +4 in NO_2 ; [1 mark]

[Total: 2 marks]

- Oxygen has an oxidation number of -2
- Overall, the compounds will have an oxidation number of 0
 - For NO , nitrogen must be +2

- For NO_2 , the two oxygen atoms have a combined oxidation number of -4 , so nitrogen must have an oxidation number of $+4$ for the overall oxidation number to be 0

3a

a) Ammonia can be described as a weak base because:

- Ammonia is a proton acceptor; [1 mark]
- That only partially dissociates / ionises in aqueous solution; [1 mark]
- $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$; [1 mark]

[Total: 3 marks]

- You must explain why ammonia is considered a base and why it is weak
- Remember:** Acids are proton donors and bases are proton acceptors
- When in an aqueous solution, ammonia reacts with water to produce ammonium ions and hydroxide ions
- The reaction is reversible and the $\rightleftharpoons$ symbol shows that not all the ammonia has

- The reaction is reversible and the $\rightleftharpoons$ symbol shows that not all the ammonia has reacted to form ammonium ions
 - You will not get the final mark unless you use this reversible symbol
- In fact, only about 1% of the ammonia will react to form ammonium ions so the equilibrium lies well over to the left-hand side

3b

b)

i) The equation for the neutralisation of aqueous ammonia by dilute sulfuric acid is:

- $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightarrow (\text{NH}_4)_2\text{SO}_4$; [1 mark]

ii) The bond angles between N-H bonds in ammonia and the ammonium ion are:

- $\text{NH}_3 = 107^\circ$
AND
 $\text{NH}_4^+ = 109.5^\circ$; [1 mark]

Explanation:

- NH_3 has 3 bonded pairs of electrons and 1 lone pair of electrons
AND
 NH_4^+ has 4 bonded pairs of electrons; [1 mark]
- The lone pair of electrons in NH_3 pushes the bonded pairs of electrons closer together; [1 mark]

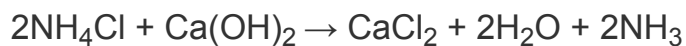
[Total: 4 marks]

- When ammonia reacts with an acid, an ammonium salt is formed
- **Remember:** Unlike typical acid-base reactions, no water is produced
- When explaining bond angles or shapes of molecules, it is always important to state the number of bonded pairs and lone pairs of electrons that the central atom has
- 4 bonded pairs of electrons will repel equally giving the tetrahedral shape with a bond angle of 109.5°
- The presence of a lone pair will reduce this angle by 2.5°

3c

c)

i) The equation for the reaction between ammonium chloride and calcium hydroxide is:



- Correct formulae: [1 mark]
- Correct balancing; [1 mark]

ii) The presence of ammonia gas can be confirmed as:

- Ammonia / it will turn damp red litmus paper blue; [1 mark]

[Total: 3 marks]

- As with any acid-base reactions, a salt and water are produced
- In addition, ammonia gas is produced when an ammonium salt is reacted with a base

